🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.NOS.6

Rational Numbers on the Number Line

Lesson 9-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can place rational numbers, including fractions and decimals, on a number line.
Language Objective: I can explain my reasoning using the words rational number, fraction, decimal, and number line.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Rational number | **Fraction** | **Decimal** | **Number line** | **Equivalent** | **Integer**

👆 *Tap any word to see what it means and a picture.*

1. Any number that can be written as a fraction, including integers and decimals, is a ____________.

2. A number that shows part of a whole, like 3/5, is a ____________.

3. A number that uses a point to show parts of a whole is a ____________.

4. A line with numbers in order used to place and compare values is a ____________.

5. Numbers that name the same value, like 1/2 and 0.5, are ____________.

6 A whole number or its opposite is an ______ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Where would you place -1 1/2 on the number line, and how do you decide between which two integers it falls?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Rational number** — Any number you can write as a fraction. This includes fractions, decimals, and integers.
  Número racional — Cualquier número que puedes escribir como fracción. Incluye fracciones, decimales y enteros.
- [ ] **Fraction** — A number that shows part of a whole, like 3/4.
  Fracción — Un número que muestra una parte de un todo, como 3/4.
- [ ] **Number line** — A straight line with numbers spaced out evenly.
  Recta numérica — Una línea recta con números separados de forma pareja.
- [ ] **Integer** — Whole numbers and their opposites, like -2, -1, 0, 1, 2.
  Número entero — Números enteros y sus opuestos, como -2, -1, 0, 1, 2.
- [ ] **Decimal** — A number with a dot, like 0.5, that shows a part less than one.
  Decimal — Un número con un punto, como 0.5, que muestra una parte menor que uno.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

-1 1/2 is between ____ and ____.
-1 1/2 esta entre ____ y ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: All of these are rational numbers because each one can be ____.

Evidence: Student explains a rational number is any number that can be written as a fraction, including integers and decimals.

Reasoning: This evidence connects the definition of rational number to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **-1 1/2 is between ____ and ____.**
  -1 1/2 esta entre ____ y ____.
- **It is closer to ____ because ____.**
  Esta mas cerca de ____ porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

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4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Rational number
2. **Notes 2:** Fraction
3. **Notes 3:** Decimal
4. **Notes 4:** Number line
5. **Notes 5:** Equivalent
6. **Notes 6:** Integer
7. **Watch for this mistake:** *A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate whole numbers instead of one value between two integers — for example, plotting -3/4 by going to -3 and then counting 4 more units to the left, landing (incorrectly) at -7, instead of recognizing that -3/4 = -0.75, which sits between 0 and -1. Before you submit an answer, find the two whole numbers your value falls between, then split that space into equal parts.*
8. **Try It 1:** A. -1.5, -0.5, 0.5 — *The farthest left on the number line comes first: -1.5, then -0.5, then 0.5.*
9. **Try It 2:** A. -1/4 — *-3/4 = -0.75 and -1/4 = -0.25. Since -0.75 is farther left (less), -3/4 is to the left of -1/4.*